

Student ID: _____

Theory Instructor Name: _____

HABIB UNIVERSITY

Digital Logic and Design - EE-172 / CS-130 / CE-222

Spring 2026



Midterm Examination

Date: 1st April 2026

Time: 75 minutes

Max points: 100

Instructions:

- i. Answer sheets should contain all steps, working, explanations, and assumptions at specified place.
- ii. Only handwritten, A4-sized two-sided, personal cheat sheet is allowed.
- iii. You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

No.	Questions	Pts
Number Systems		
Q1a)	Convert the number $(235)_7$ into: i) Decimal ii) Binary iii) Hexadecimal	6
A1	i) $= 2(7^2) + 3(7^1) + 5(7^0)$ $= 2(49) + 3(7) + 5$ $= 98 + 21 + 5 = 124_{10}$ ii) $124_{10} = 1111100_2$ iii) $0111\ 1100 = 7C_{16}$	
Q1b)	Perform the following addition in BCD and show all correction steps: $472_{10} + 389_{10}$	4
A	$472 =$ $4 \rightarrow 0100$ $7 \rightarrow 0111$ $2 \rightarrow 0010$ $389 =$ $3 \rightarrow 0011$ $8 \rightarrow 1000$ $9 \rightarrow 1001$ $\begin{array}{r} 11 \\ 0100\ 0111\ 0010 \\ + 0011\ 1000\ 1001 \\ \hline = 1000\ 0000\ 1011 \\ 0110\ 0110 \\ \hline = 1000\ 0110\ 0001 \end{array}$ $472 + 389 = 861$	

Q1c)	A small delivery drone records its vertical movement relative to its starting position. - In the first interval, it moves upward by +22 meters. - In the second interval, it descends by 37 meters. Upward movement is considered positive, and downward movement is negative. - Represent both movements in signed 2's complement form using the minimum number of bits required to represent both values. - Determine the drone's net displacement using binary 2's complement addition. Show your working clearly and state whether overflow occurs. - Convert the final result back to decimal and interpret the physical meaning.	2+ 2+ 2
	i) Need 7 bits (range -64 to +63) $+22_{10} = 0010110_2$ $37_{10} = 0100101_2$ $-37_{10} = 1011011_2$ ii) 0010110 +1011011 ----- 1110001 Signs different → No overflow 1110001 is negative. iii) 2's complement: $0001111 = 15$ So, $= -15$ Net displacement = -15 meters Drone is 15 meters below starting position.	
Q1d)	State whether the following are True or False. Justify briefly if incorrect. 7 is the minimum number of bits required to represent -90 in signed 2's complement. False. 7 bits → -64 to +63 (not enough) 8 bits → -128 to +127 (sufficient) Minimum = 8 bits - Overflow occurs when adding two positive 2's complement numbers and the result is negative. True - BCD addition requires correction if the 4-bit sum is greater than 1001. True	1+2+1

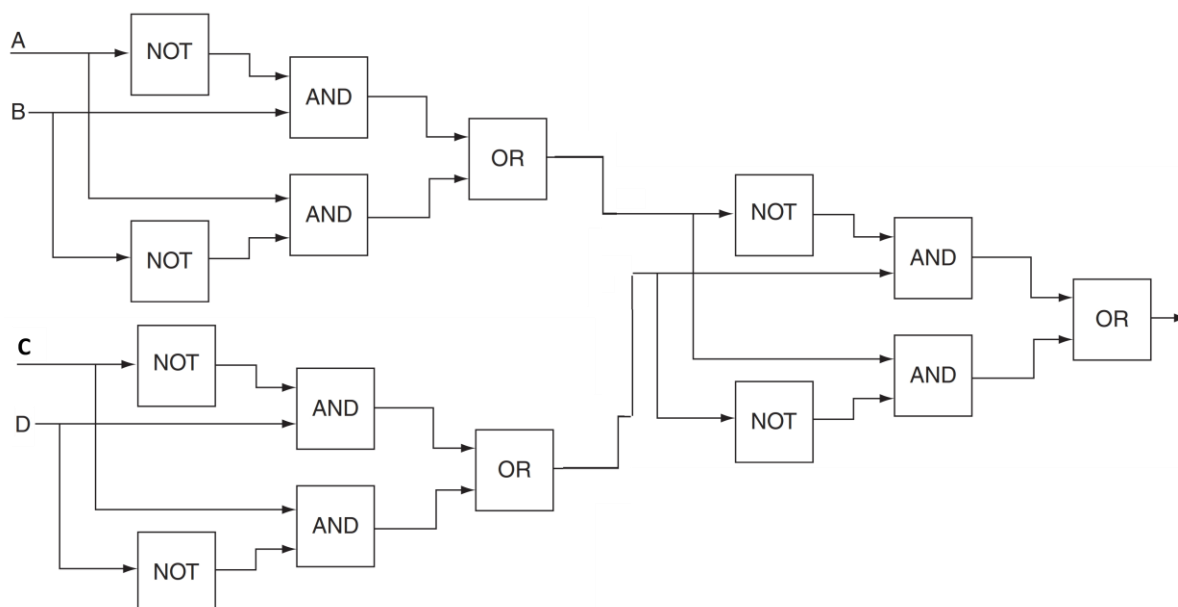
Boolean Algebra and equivalent realizations

Q2a

Implement a 4-input XOR function using AND, OR and NOT gates. Assume inputs as, A, B, C, D.

7

A



Q2b

There are four switches (S1, S2, S3 and S4) that are part of the control circuitry in a copying machine. The switches are at various points along the path of the copy paper as the paper passes through the machine. Each switch is normally open (logic 0), and as the paper passes over the switch, the switch closes (logic 1). It is impossible for the switches S1 and S4 to be closed at the same time. Design a simplified logic circuit to produce a HIGH output whenever three or more switches are closed at the same time.

- Draw the truth table using S1, S2, S3, S4 as input variables and F as output variable.
- Write the expression for F in canonical SOP form, showing the relevant literals.
- Using K-map, write down the simplified expressions for each output.

3+3+7

Ans	i.	<table><tr><th>S1</th><th>S2</th><th>S3</th><th>S4</th><th>F</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>X</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>X</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>X</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>X</td></tr></table>	S1	S2	S3	S4	F	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	1	1	0	0	1	0	0	0	0	1	0	1	0	0	1	1	0	0	0	1	1	1	1	1	0	0	0	0	1	0	0	1	X	1	0	1	0	0	1	0	1	1	X	1	1	0	0	0	1	1	0	1	X	1	1	1	0	1	1	1	1	1	X	
	S1	S2	S3	S4	F																																																																																			
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	ii.	0 0 0 0 0 0 1 0 0 X X 1 0 X X 0 iii. $F = S1.S2.S3 + S2.S3.S4$																																																																																						
Q2c	Simplify and find the complement of the following expression and express the complement in canonical POS form: $F(x, y, z) = y.(x'z' + xy'z)$ $G(x, y, z) = xyz + xy'$	5+10																																																																																						
Ans	$F = x' . y . z'$ $F' = (x + y' + z)$ $F = \sum(2), F' = \prod(2)$ $G \text{ (already simplified)} = xyz + xy'z' + xy'z$ $G = \sum(4,5,7), G' = \prod(4,5,7)$																																																																																							
Combinational Logic																																																																																								
Q3	A digital system has four input lines A, B, C, D, where the 4-inputs represent a 4-bit binary number (0 – 15). The system, based on the 4-inputs, produces a 3-bit output Y2, Y1, Y0 and a control signal V (see below). The system operates as follows:	10+4+6																																																																																						

- 1) The input (A, B, C, D) represents a 4-bit binary number (0 - 15) where input line, A represents the Most Significant Bit (MSB).
- 2) When the input is in the range 0 – 7, the control signal output $V = 1$ and when the input is in the range 8 – 15, the control signal output $V = 0$.
- 3) If the inputs correspond to the binary number 0 - 7, the output Y_2, Y_1, Y_0 is equal to the input binary value (same as input).
- 4) If the inputs correspond to the binary number 8 - 15, the output Y_2, Y_1, Y_0 is the binary equivalent of (Input – 8). Note – this is 8 subtracted from the binary number (0 – 15).
- We will now implement the system using a 4x16 Decoder (active-high outputs) and OR gates. Implement and show the circuit, using only the outputs ($I_0, I_1, \dots, I_{15}$) of the 4x16 Decoder and OR gates for the following:
- (a) Capture the system using an appropriate Truth Table.
- (b) Control signal V.
- (c) Outputs, Y_2, Y_1, Y_0 .

Ans

Inputs				Control Signal	Outputs		
A	B	C	D	V	Y_2	Y_1	Y_0
0	0	0	0	1	0	0	0
0	0	0	1	1	0	0	1
0	0	1	0	1	0	1	0
0	0	1	1	1	0	1	1
0	1	0	0	1	1	0	0
0	1	0	1	1	1	0	1
0	1	1	0	1	1	1	0
0	1	1	1	1	1	1	1
1	0	0	0	0	0	0	0
1	0	0	1	0	0	0	1
1	0	1	0	0	0	1	0
1	0	1	1	0	0	1	1
1	1	0	0	0	1	0	0
1	1	0	1	0	1	0	1
1	1	1	0	0	1	1	0
1	1	1	1	0	1	1	1

The first 8 outputs of the 4x16 Decoder (I_0 to I_7) are OR'ed using an 8-input OR gate to implement V.

(a) [6 points] We see that $Y_0 = D$, $Y_1 = C$ and $Y_2 = B$

Implementing Y_0

- 2) Use the 8 outputs of the Decoder, $I_1, I_3, I_5, I_7, I_9, I_{11}, I_{13}$ and I_{15} and an 8-input OR gate to implement Y_0

Implementing Y_1

- 2) Use the 8 outputs of the Decoder, $I_2, I_3, I_6, I_7, I_{10}, I_{11}, I_{14}$ and I_{15} and an 8-input OR gate to implement Y_1

Implementing Y_2

	1) Use the 8 outputs of the Decoder, $I_4, I_5, I_6, I_7, I_{12}, I_{13}, I_{14}$ and I_{15} and an 8-input OR gate to implement Y_2																																									
Q4	a) In a robotic navigation system, there are four sensors (S_3, S_2, S_1, and S_0) that detect obstacles at different distances. Each sensor sends a signal if an obstacle is detected within its range, with S_3 being the sensor that detects the closest obstacles and S_0 the farthest. As such, the robot's control system also prioritizes obstacles based on proximity, with S_3 having the highest priority and S_0 the lowest. Design a priority encoder that takes inputs from these four sensors and outputs a 2-bit code (P_1, P_0) indicating the highest-priority sensor that detects an obstacle. <table><tr><th>Priority Sensor Detected</th><th>P_1</th><th>P_0</th></tr><tr><td>S_0</td><td>0</td><td>0</td></tr><tr><td>S_1</td><td>0</td><td>1</td></tr><tr><td>S_2</td><td>1</td><td>0</td></tr><tr><td>S_3</td><td>1</td><td>1</td></tr></table> b) Design a 2-to-4 decoder using two 1-to-2 DeMux (with enable inputs). c) The encoded data must be decoded for the device that controls the robot's movement. Based on the previous priority encoder outputs (P_1 and P_0), evasive maneuvers are defined by output signals M (movement) and T (turn), which control the robot's movement response. $M = 1$ means the robot will stop, and $T = 1$ means the robot will turn. <table><tr><th>Condition</th><th>P_1</th><th>P_0</th><th>M</th><th>T</th></tr><tr><td>Obstacle closest</td><td>1</td><td>1</td><td>1</td><td>0</td></tr><tr><td>Obstacle near</td><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>Obstacle far</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>Obstacle furthest</td><td>0</td><td>0</td><td>0</td><td>1</td></tr></table> Design this combinational logic circuit using two 1-To-2 demultiplexers to determine which maneuvers the robot should take.	Priority Sensor Detected	P_1	P_0	S_0	0	0	S_1	0	1	S_2	1	0	S_3	1	1	Condition	P_1	P_0	M	T	Obstacle closest	1	1	1	0	Obstacle near	1	0	0	1	Obstacle far	0	1	0	1	Obstacle furthest	0	0	0	1	10+ 5 + 10
Priority Sensor Detected	P_1	P_0																																								
S_0	0	0																																								
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S_2	1	0																																								
S_3	1	1																																								
Condition	P_1	P_0	M	T																																						
Obstacle closest	1	1	1	0																																						
Obstacle near	1	0	0	1																																						
Obstacle far	0	1	0	1																																						
Obstacle furthest	0	0	0	1																																						

Ans

	S ₃	S ₂	S ₁	S ₀	P ₁	P ₀
No input	0	0	0	0	X	X
	0	0	0	1	0	0
	0	0	1	X	0	1
	0	1	X	X	1	0
	1	X	X	X	1	1

wx\yz	00	01	11	10
00				
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$P_1 = w + x$

wx\yz	00	01	11	10
00			1	1
01				
11	1	1	1	1
10	1	1	1	1

$P_0 = w + x'y$

